

Abubakar Shaikh

✉ abu2012only@gmail.com ☎ +1 825 523-2882 🌐 github/Akuma277353 🌐 linkedin/abu-shaikh 📁 portfolio

EDUCATION

University of Alberta

09/2023 – Present | Edmonton, AB

Bachelor of Science in Computing Science (Major in A.I)

- Relevant Coursework: Algorithms, Computer Architecture, Reinforcement Learning, Machine Learning, Software Engineering, File & DBMS, Web Apps & Architecture, Search & Planning in AI
- Expected Graduation: 2027

PROJECTS

Autonomous Overload Remediation System (AORS)

01/2026 – Present

Python | FastAPI | scikit-learn | psutil | Streamlit

- Demonstrated recovery from **sustained 90%+ CPU saturation in <30s** by building a closed-loop controller on **real hardware** with psutil telemetry + temporal signal fusion.
- Eliminated hardcoded thresholds for work-intake scaling by implementing online **adaptive control (EWMA & Welford)** to map request rate to CPU utilization in real time.
- Enabled targeted mitigation of traffic anomalies by training a scikit-learn classifier to separate normal spikes from retry storms.
- Shipped a production-style control surface via a decoupled **FastAPI** control plane and **Streamlit** dashboard for monitoring and interventions.

Data-Driven Event Allocation Platform (AuroraChance)

10/2025 – 12/2025

Android | Firebase | Java | Google Maps SDK

- Built a real-time event lottery platform on **Firebase Firestore**, designing structured collections for **events, users, waitlists, and notifications** with multi-role sync (entrants/organizers/admins) under concurrent updates.
- Added **location-based validation** by integrating **Google Maps + Places APIs** for geolocation capture and checks.
- **Delivered 45+ user stories in 9 weeks** by coordinating a 6-member team with GitHub workflows and Agile Kanban; improved reliability using UML/CRC and automated tests.

Mini Compiler Back End: RISC-V to WebAssembly

10/2025 – 11/2025

RISC-V | WASM | LEB128

- Built a **binary translator from RISC-V to WASM** bytecode while preserving program semantics.
- Implemented correct instruction lowering by mapping I-type, R-type, and branch operations into WASM expressions (immediates + zero register handling).
- Generated compact WASM immediates by implementing **LEB128** encoding with correct byte counts.
- Supported structured control flow by computing forward/backward branch targets and emitting correct block/loop/end bytecode using a target-count table.

LEADERSHIP AND AWARDS

Awards & Recognition

04/2023

- **3rd Place**, World Robotics Olympiad Nationals UAE
- University of Alberta Regional Excellence Scholarship – **\$5,000**
- University of Alberta International Admission Scholarship - **\$5000**

Good Will Ambassador's Club, President

05/2022

- Organized and ran multiple charity drives and fundraising events as club President.
- Kept initiatives on track through weekly meetings, timeline alignment, and supporting long-term club growth.

SKILLS

Programming languages

Python, Bash, C, Java, JavaScript, SQL, RISC-V Assembly

Web & App Dev

HTML, CSS, RESTful APIs, FastAPI, Streamlit, Android Development

Cloud & DevOps

AWS (Solutions Architect Associate – in progress, completing Feb 2026), CI/CD fundamentals

Software Development

Object-Oriented Programming, DSA, Version Control (Git/GitHub), Agile/Kanban

Tools and Frameworks

Linux, VS Code, Android Studio, Firebase, React, MongoDB, Django, Heroku, npm

Data Science & ML

NumPy, Pandas, Matplotlib, scikit-learn, time-series analysis, data QC/outlier handling